

Ahmad Rammal

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EDUCATION

MVA (Mathématiques, Vision, Apprentissage) - Master 2 2023 – 2024

Highest Honors, GPA: 4/4

- *Highly* competitive renowned program in the field of mathematical modelling and machine learning
- Relevant Coursework: Reinforcement Learning, Advanced learning for text and graph data, AGI Safety, Distributed and Large Scale Computing Optimization, Deep Learning in practice.

ECOLE POLYTECHNIQUE - Ingénieur Polytechnicien

2020 – 2023

High Distinction, GPA: 3.91/4

- Merit scholarship for my 3rd -year academic performance (**Year GPA: 3.96/4**)
- Specialization in Applied Mathematics

SAINT JOSEPH UNIVERSITY - French preparatory classes MP*

2018 – 2020

Ranked among top 1% in the faculty of engineering.

RESEARCH EXPERIENCE

Meta Fundamental AI Research (FAIR)

Paris, France

AI Research Assistant (PhD)

October 2025 – Present

Research on self-improving AI systems for formal mathematical reasoning under the supervision of Rémi Munos and Amaury Hayat.

BNP Paribas – External Consultant

Paris, France

AI Researcher

July 2025 – October 2025

Research on RAG-based pipelines to parse and structure bank documents, and support quicker decisions in credit application reviews.

JPMorgan Chase & co

Paris, France

AI Researcher Intern

April 2024 – October 2024

- Research on Mathematical Reasoning in LLMs:
 - Received outstanding paper award at *NeurIPS 2024 Workshop MATH-AI* (top 4%). Available on [arXiv:2410.16973](https://arxiv.org/abs/2410.16973).
 - Evaluated the impact of synthetic data on improving mathematical problem-solving abilities.
 - Enhanced LLM architecture for better performance in math tasks.

King Abdullah University of Science and Technology

Thuwal, Saudi Arabia

Researcher in the group of Peter Richtárik

April 2023 – October 2023

Nominated by Ecole Polytechnique for the best internship award.

Contributed to two research papers:

- *Communication Compression for Byzantine Robust Learning*:
 - Paper accepted at *AISTATS 2024* and available on arXiv: [arXiv:2310.09804](https://arxiv.org/abs/2310.09804)
 - Led the research effort as first author.
- *Correlated Quantization for Faster Non-convex Distributed Optimization*:
 - Selected for oral presentation at *UAI 2025* and available on arXiv: [arXiv:2401.05518](https://arxiv.org/abs/2401.05518)

PROJECTS & OPEN-SOURCE CONTRIBUTIONS

Pruned GPT

February 2025

Designed a compact GPT-2 model between GPT-2 Small and Medium by implementing NVIDIA's pruning and knowledge distillation approach in [arXiv:2407.14679](https://arxiv.org/abs/2407.14679)

Gram-Schmidt Model

June 2022 – September 2022

A semi-supervised approach applied on binary classification based on the theory of orthonormal spaces.

SKILLS AND HOBBIES

- **Programming** – Python, Pytorch, *Scikit-Learn*, *Flask*, *Git*, *Aws*
- **Languages** - English fluent, French fluent, Arabic native, German beginner.
- **Hobbies** – Calisthenics, climbing, guitar, cycling, reading.